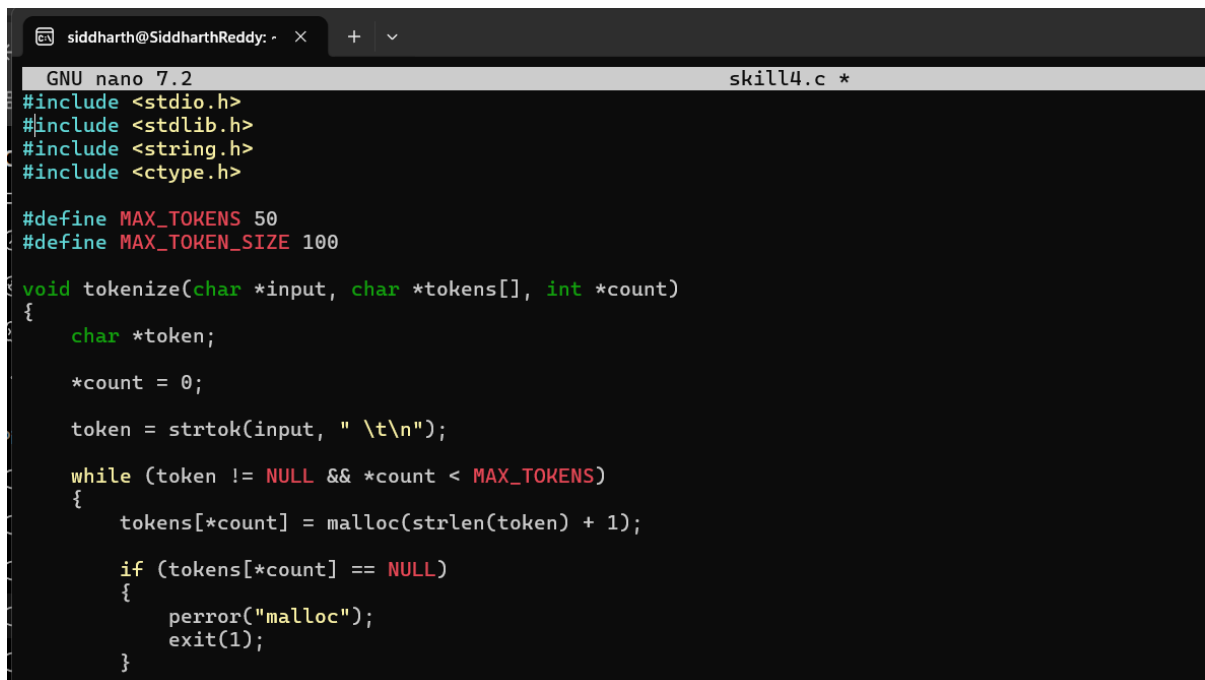


Skill -4 OS



```
GNU nano 7.2 skill4.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>

#define MAX_TOKENS 50
#define MAX_TOKEN_SIZE 100

void tokenize(char *input, char *tokens[], int *count)
{
    char *token;

    *count = 0;

    token = strtok(input, " \\t\\n");

    while (token != NULL && *count < MAX_TOKENS)
    {
        tokens[*count] = malloc(strlen(token) + 1);

        if (tokens[*count] == NULL)
        {
            perror("malloc");
            exit(1);
        }
    }
}
```

Skill -4 OS

```
siddharth@SiddharthReddy: ~  
GNU nano 7.2 skill4  
    }  
  
    strcpy(tokens[*count], token);  
  
    (*count)++;  
  
    token = strtok(NULL, " \\t\\n");  
}  
}  
  
void free_tokens(char *tokens[], int count)  
{  
    for (int i = 0; i < count; i++)  
        free(tokens[i]);  
}  
  
void parse_tokens(char *tokens[], int count)  
{  
    if (count == 0)  
    {  
        printf("Empty command\\n");  
        return;  
    }  
  
    printf("\\nParser Output:\\n");  
    printf("Command: %s\\n", tokens[0]);  
}
```

Skill -4 OS

```
siddharth@SiddharthReddy: ~  
GNU nano 7.2 skill4.c  
  
    printf("Arguments:\n");  
  
    for (int i = 1; i < count; i++)  
    {  
        printf("  Argument %d: %s\n", i, tokens[i]);  
    }  
}  
  
int main()  
{  
    char input[500];  
    char *tokens[MAX_TOKENS];  
    int count;  
  
    while (1)  
    {  
        printf("> ");  
  
        if (fgets(input, sizeof(input), stdin) == NULL)  
            break;  
  
        input[strcspn(input, "\n")] = '\0';  
  
        if (strcmp(input, "exit") == 0)  
            break;  
    }  
}
```

Skill -4 OS

```
siddharth@SiddharthReddy: ~ × + v
GNU nano 7.2 skill4.c *
    input[strcspn(input, "\n")] = '\0';

    if (strcmp(input, "exit") == 0)
        break;

    tokenize(input, tokens, &count);

    printf("\nTokens:\n");

    for (int i = 0; i < count; i++)
    {
        printf("Token %d: %s\n", i + 1, tokens[i]);
    }

    parse_tokens(tokens, count);

    free_tokens(tokens, count);

    printf("\n");
}

printf("Exiting Skill 4...\n");

return 0;
}
```

```
siddharth@SiddharthReddy:~/ossp_practical$ nano skill4.c
siddharth@SiddharthReddy:~/ossp_practical$ gcc skill4.c -o skill4
siddharth@SiddharthReddy:~/ossp_practical$ ./skill4
> ls -l

Tokens:
Token 1: ls
Token 2: -l

Parser Output:
Command: ls
Arguments:
    Argument 1: -l

> |
```

Skill -4 OS

```
>
```

```
Tokens:
```

```
Empty command
```

```
> |
```